Function	Description
Dynamic masking	Blocking a part of the video signal based on the signal itself, for example, because of privacy concerns.
Flame and smoke detection	IP cameras with intelligent video surveillance technology are being used to detect flames and smoke within 15 to 20 seconds or even less because of the built-in DSP chip.
Egomotion estimation	This is used to determine the location of a camera by analysing its output signal.
Motion detection	This determines the presence of relevant motion in the observed scene.
Shape recognition	This recognises shapes in the input video, for example, circles or squares. This feature is one aspect of more advanced functionalities such as object detection.
Object detection	Object detection determines the presence of a type of object or entity, for example a person or car.
Recognition	Face recognition and automatic number plate recognition make it possible to identify individuals or cars.
Style detection	Style detection is used in settings where a video signal has been produced, for example, for television broadcast.
Tamper detection	Tamper detection tells us whether the camera or output signal has been tampered with.
Video tracking	This feature is used to determine the location of persons or objects in the video signal, possibly with regard to an external reference grid.
Object co-segmentation	This involves the joint object discovery, classification and segmentation of targets in one or multiple related video sequences.

Table 1: Video analytics functionalities (courtesy Wikipedia)

TensorFlow is an open source software library for numerical computation using data flow graphs. The graph nodes represent mathematical operations, while the graph edges represent the multi-dimensional data arrays (tensors) that flow between them. This flexible architecture enables users to deploy computation to one or more CPUs or GPUs in a desktop, server, or mobile device without rewriting code. TensorFlow is an end-to-end open source platform for machine learning. It has a comprehensive, flexible ecosystem of tools, libraries and community resources that lets researchers push the limits of the state-of-art in ML, and helps developers easily build and deploy ML-powered applications.

Deep learning is a part of machine learning in artificial intelligence. It works in a manner that is similar to how the human brain works to process data and to create patterns, which are used for decision making. Deep learning allows computational models to learn representation of data with multiple levels of abstraction. These

```
File Edit View Search Terminal Help
surabhi@surabhi-seng:-$ pip install tensorflow
Collecting tensorflow
  Downloading https://files.pythonhosted.org/packages/d3/59/d88fe8c58ffb66aca21d03c0e290cd68327cc133591130c674985e98a482/tensorf
low-1.14.0-cp27-cp27mu-manylinux1_x86_64.whl (109.2MB)
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Collecting grpcio>=1.8.6 (from tensorflow)
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1.22.0-cp27-cp27mu-manylinux1_x86_64.whl (2.2MB)
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Collecting mock>=2.0.0 (from tensorflow)
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0.5-py2.py3-none-any.whl
Collecting keras-applications>=1.0.6 (from tensorflow)
  Downloading https://files.pythonhosted.org/packages/21/56/4bcec5a8d9503a87e58e814c4e32ac2b32c37c685672c30bc8c54c6e478a/Keras_A
pplications-1.0.8.tar.gz (289kB)
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Collecting wrapt>=1.11.1 (from tensorflow)
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.11.2.tar.gz
Collecting protobuf>=3.6.1 (from tensorflow)
  Downloading https://files.pythonhosted.org/packages/30/bf/d7f8bc958f9eb4661498cde9f7e630da863e43278222d46105712dde4e13/protobu
f-3.9.0-cp27-cp27mu-manylinux1_x86_64.whl (1.2MB)
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Collecting keras-preprocessing>=1.0.5 (from tensorflow)
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reprocessing-1.1.0-py2.py3-none-any.whl (41kB)
    100% |████████████████████████████████| 51kB 6.7MB/s
Collecting gast>=0.2.0 (from tensorflow)
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2.2.tar.gz
Collecting wheel (from tensorflow)
  Downloading https://files.pythonhosted.org/packages/bb/10/44230dd6bf3563b8f227dbf344c908d412ad2ff48066476672f3a72e174e/wheel-0
.33.4-py2.py3-none-any.whl
Collecting tensorboard<1.15.0,>=1.14.0 (from tensorflow)
  Downloading https://files.pythonhosted.org/packages/f4/37/e6a7af1c92c5b68fb427f853b06164b56ea92126bcfd87784334ec5e4d42/tensorb
oard-1.14.0-py2-none-any.whl (3.1MB)
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Figure 1: TensorFlow installation